SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4_AT2 — VISUALIZATION EVALUATION EXERCISE

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO4 – Evaluation (BL5)	**Assessment:**	Assessment Tool 2 – Visualization Evaluation Exercise
Weightage:	20% of CIA	**Total Marks:**	25 (5 Questions × 5 Marks)
CO4 Statement:	*Evaluate the appropriateness and design choices of relationship-visualization techniques (scatterplots, correlograms, PCA, bubble/hexbin plots, and network graphs) and justify improvements.(BL5)*		
Student Name:	Sania Herbert	**Reg. No.:**	192324062
Section / Batch:	2023/AI&DS	**Date:**	23.08.26

Code of Conduct

I Sania Herbert(192324062) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sania

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond describing the scenario: evaluate the specific strength or weakness, and justify your judgment.
- Where a question asks for a recommendation, propose a specific alternative — not just “use a different chart.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – VISUALIZATION EVALUATION EXERCISE

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT2 — Visualization Evaluation Exercise

CO4 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO4-AT2 Visualization Evaluation Exercise problems, in which students are given a visualization design choice and must evaluate its appropriateness, identify limitations, and justify improvements.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough understanding of the relevant visualization principle being evaluated (e.g., overplotting, PCA variance, encoding choice, network readability).	Good understanding of the relevant principle, with only minor gaps.	Basic understanding, but with some misconceptions about which principle applies.	Poor understanding of core principles; misidentifies the nature of the issue entirely.
Explanation	**25%**	Clearly explains why the design choice is effective or ineffective, including its practical consequence for a reader — not just that it is ‘good’ or ‘bad.’	Explains the evaluation with adequate reasoning, covering most of the practical consequence.	Identifies the issue but explanation is generic or only loosely tied to its practical effect.	Little or no explanation provided; the evaluation is a bare assertion.
Application	**20%**	Proposes a specific, well-justified recommendation or alternative that directly addresses the	Proposes a workable recommendation, with only minor gaps in justification.	Proposes a recommendation, but it only partially addresses the issue or lacks justification.	Fails to propose a workable recommendation, or it does not address the evaluated issue.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		evaluated issue.			
Organization	**15%**	The response is clearly structured, addressing evaluation and recommendation in a logical sequence with coherent overall flow.	The response is organized, with only minor issues in flow or clarity.	Basic structure; both parts are addressed but the response lacks clarity or sequence.	Poorly organized; the response is incoherent, and one or both parts are left unaddressed.
Accuracy	**10%**	Both the evaluation and the recommendation are completely accurate and well-suited to the specific scenario described.	Mostly accurate, with only minor omissions or a small error.	Partially correct; the evaluation or recommendation contains notable inaccuracies.	Largely incorrect or inappropriate, with major gaps in both the evaluation and recommendation.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Problem 5
Data: A scatterplot shows a strong visual pattern, and the analyst concludes that X directly causes Y
based solely on this chart.
Task: Evaluate this conclusion and identify what additional evidence would be needed to support it.

A scatterplot showing a strong visual pattern indicates that there may be a **relationship or correlation** between X and Y. However, it is incorrect to conclude that **X directly causes Y** based only on the scatterplot.

Correlation does not imply causation because:

- A **third variable (confounding variable)** may affect both X and Y.
- The direction of the relationship may be reversed.
- The observed relationship may occur by chance.
- Other uncontrolled factors may influence Y.

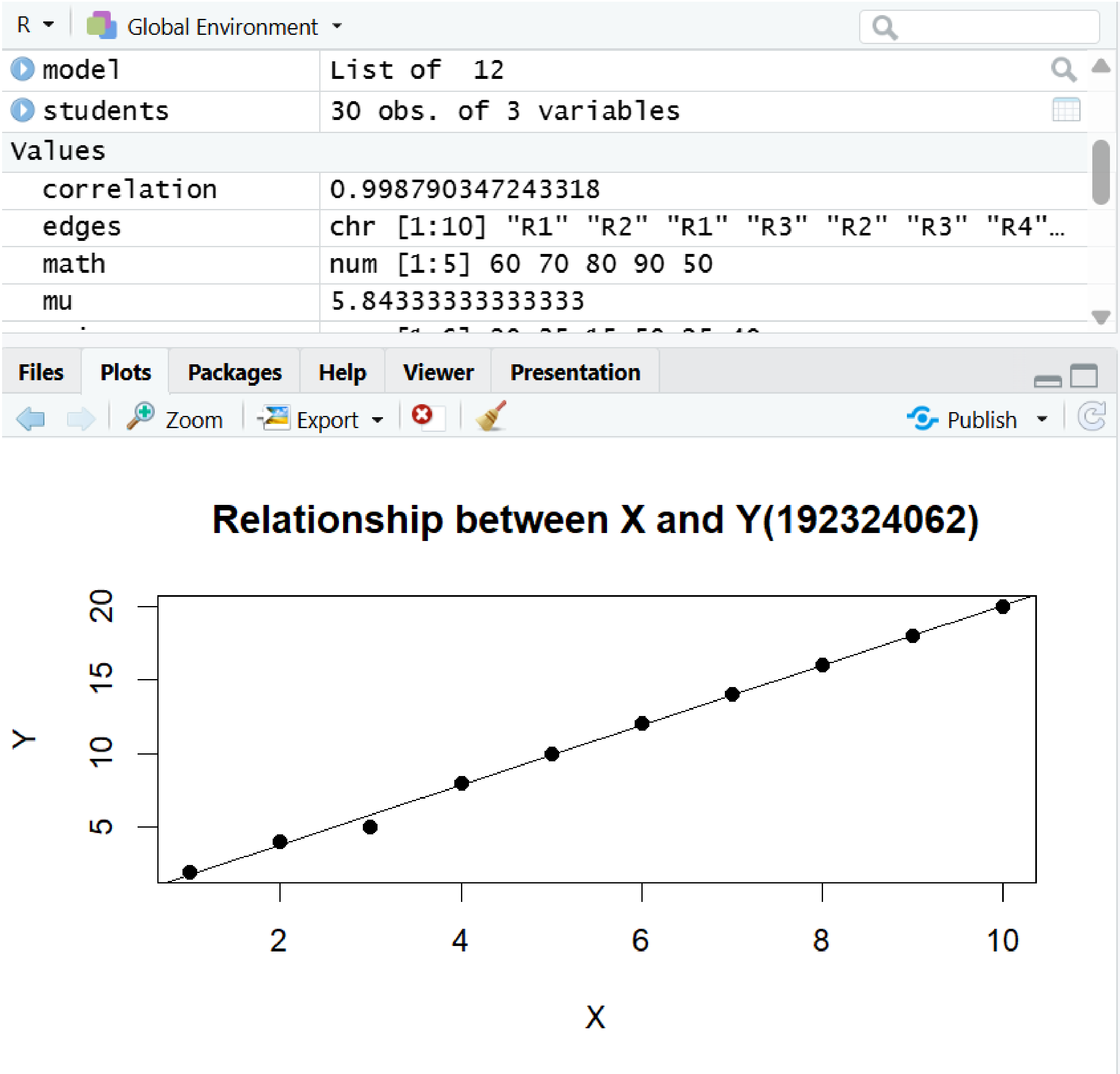
To support a causal relationship, we need **controlled experiments**, evidence that **X occurs before Y**, control of confounding variables, statistical significance tests, and replication of the result.

```r
# Create sample data
X <- c(1, 2, 3, 4, 5, 6, 7, 8, 9, 10)
Y <- c(2, 4, 5, 8, 10, 12, 14, 16, 18, 20)

# Create scatterplot
plot(X, Y,
     main = "Relationship between X and Y(192324062)",
     xlab = "X",
     ylab = "Y",
     pch = 19)

# Calculate correlation
correlation <- cor(X, Y)
print(paste("Correlation =", correlation))

# Fit linear regression
model <- lm(Y ~ X)
```

23:14 (Top Level) R Script

R 4.6.1 · ~/

```
(Intercept) -0.26667    0.21927  -1.216    0.259
X            2.03030    0.03534  57.452 9.35e-12 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.321 on 8 degrees of freedom
Multiple R-squared:  0.9976,    Adjusted R-squared:  0.9973
F-statistic:  3301 on 1 and 8 DF,  p-value: 9.354e-12

>
> # Add regression line
> abline(model)
>
```



Interpretation

The scatterplot and high correlation indicate a **strong association between X and Y**. The regression model shows how Y changes with X. However, this code **does not prove that X causes Y**.

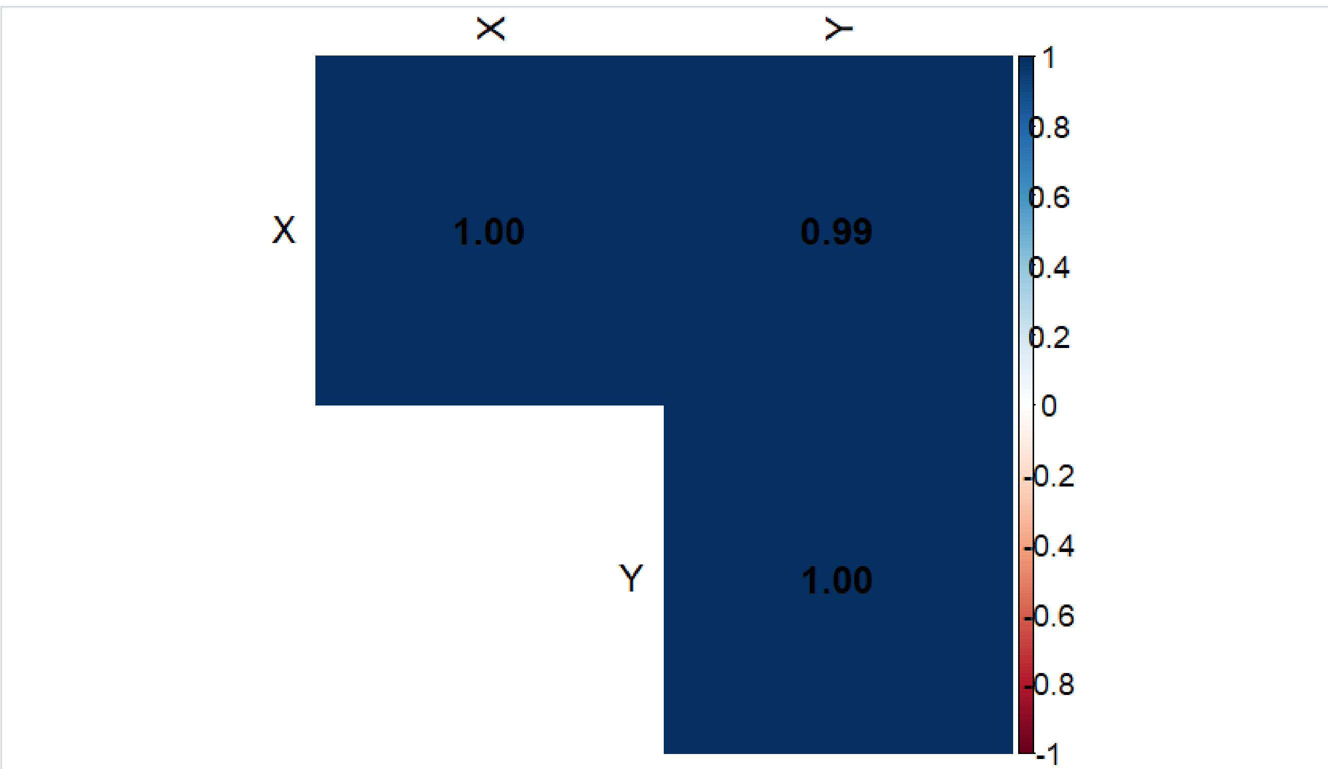
To establish causation, we would need experimental or additional statistical evidence controlling for other possible factors.

Problem 6

Data: A correlogram is built using only 2 variables, producing a single off-diagonal cell.

Task: Evaluate whether a correlogram is the right tool for comparing just 2 variables.

```r
# Install once if needed
# install.packages("corrplot")

library(corrplot)

X <- c(10,20,30,40,50,60,70,80)
Y <- c(12,25,28,45,48,65,68,82)

# Create correlation matrix
data <- data.frame(X, Y)
cor_matrix <- cor(data)

# Display correlation matrix
print(cor_matrix)

# Create correlogram
corrplot(cor_matrix,
```



Problem 12

Data: An analyst keeps only PC1 (explaining 40% of total variance) for a final 2D visualization requirement.

Task: Evaluate whether this decision is justified, and what you would recommend instead.

Keeping only **PC1**, which explains **40% of the total variance**, is **not fully justified** when the requirement is to create a **2D visualization**.
PC1 contains only 40% of the information/variance in the original dataset. The remaining **60% of the variance** is lost if PC1 alone is used.
For a 2D visualization, we should use **PC1 and PC2**. This allows us to visualize the data in two dimensions while retaining more information.
For example, if PC2 explains 25% of the variance:

- PC1 = 40%
- PC2 = 25%
- **Total = 65%**

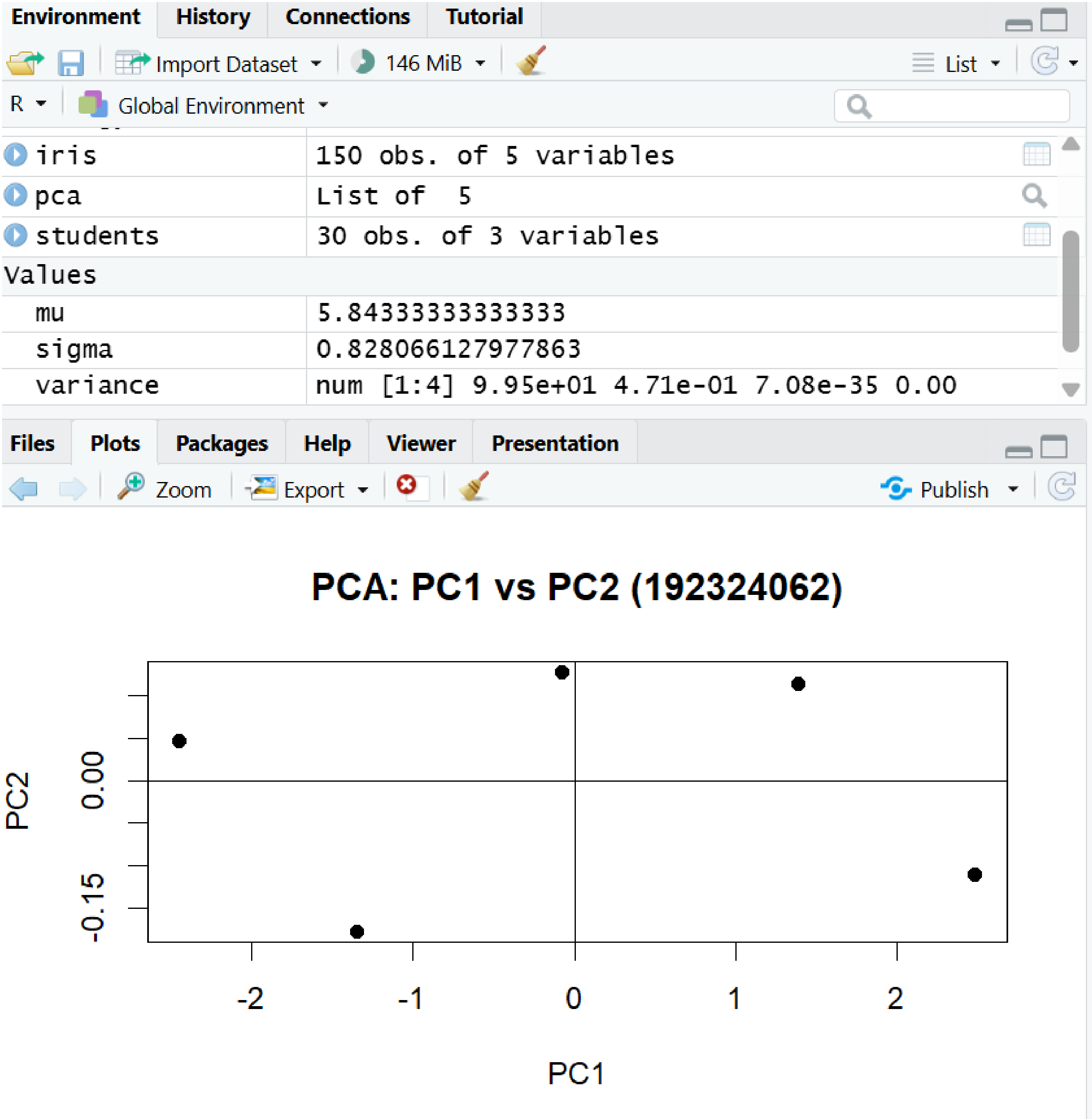
Thus, PC1 + PC2 gives a much better representation than PC1 alone.

```r
# Display PCA summary
summary(pca)

# Variance explained
variance <- pca$sdev^2 / sum(pca$sdev^2) * 100
print(variance)

# 2D visualization using PC1 and PC2
plot(pca$x[,1], pca$x[,2],
     main = "PCA: PC1 vs PC2 (192324062)",
     xlab = "PC1",
     ylab = "PC2",
     pch = 19)

# Add reference lines
abline(h = 0)
abline(v = 0)
```

28:14 (Top Level) R Script

R 4.6.1 · ~/

```
[1] 9.952941e+01 4.705868e-01 7.083151e-35 0.000000e+00
>
> # 2D visualization using PC1 and PC2
> plot(pca$x[,1], pca$x[,2],
+      main = "PCA: PC1 vs PC2 (192324062)",
+      xlab = "PC1",
+      ylab = "PC2",
+      pch = 19)
>
> # Add reference lines
> abline(h = 0)
> abline(v = 0)
>
```



Problem 18

Data: A hexbin plot is used to visualize a dataset containing only 50 data points.

Task: Evaluate whether hexbin binning was necessary or appropriate for this dataset size.

A hexbin plot divides the plotting area into hexagonal bins and counts how many observations fall into each bin. It is mainly useful for large datasets, where thousands of points overlap and a normal scatterplot becomes difficult to interpret.

With only 50 data points, hexbin binning is generally not necessary. There are relatively few observations, so a normal scatterplot can display the individual points clearly.

Using a hexbin plot with only 50 observations may:

- Hide individual data points.
- Add unnecessary complexity.
- Provide little advantage over a scatterplot.
- Make the distribution harder to interpret.

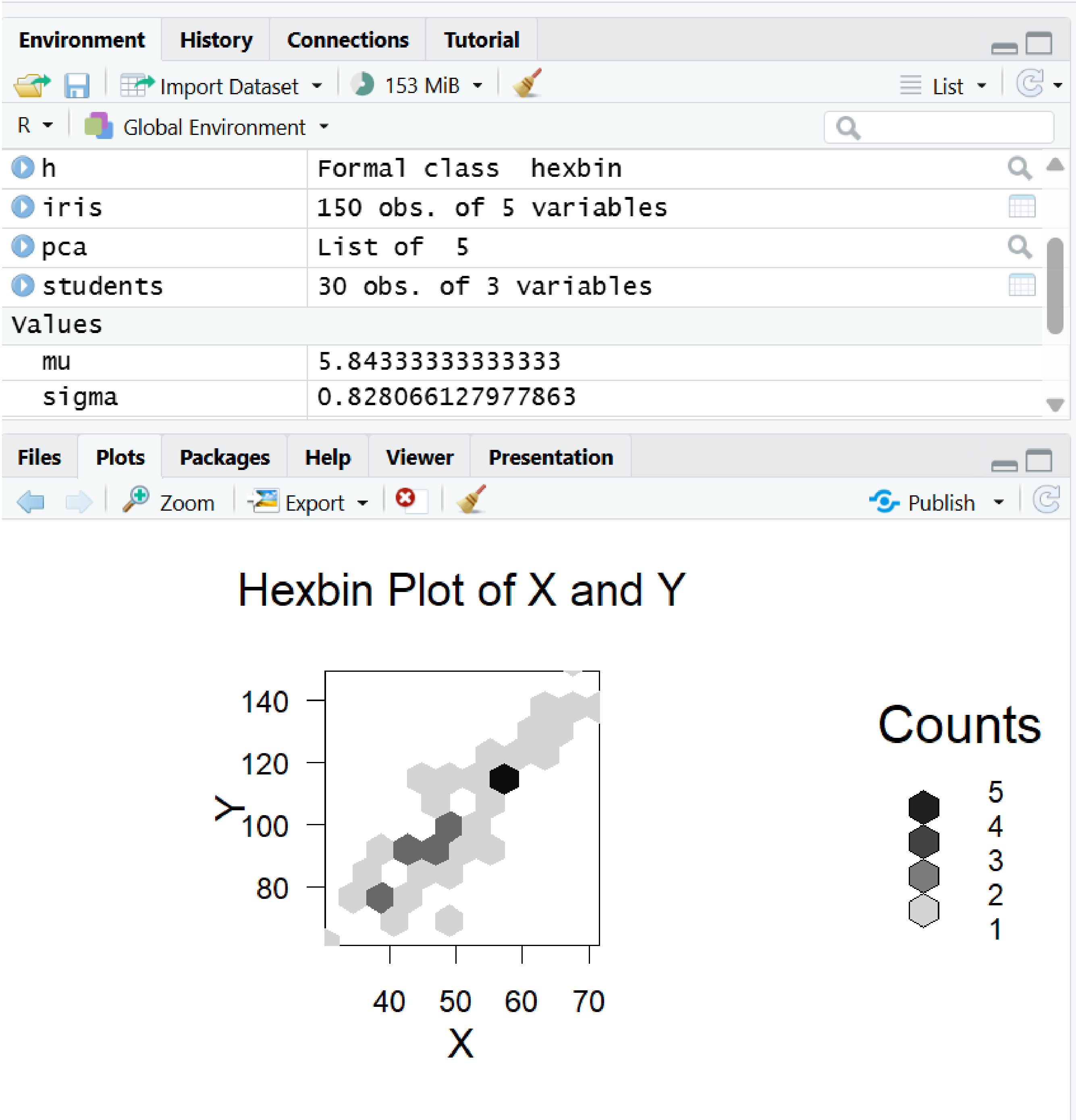
Therefore, a scatterplot is more appropriate for this small dataset.

```r

library(hexbin)

set.seed(123)

# Create 50 data points
X <- rnorm(50, mean = 50, sd = 10)
Y <- 2 * X + rnorm(50, mean = 0, sd = 10)

# Create hexbin object
h <- hexbin(X, Y, xbins = 10)

# Plot hexbin
plot(h,
     main = "Hexbin Plot of X and Y",
     xlab = "X",
     ylab = "Y")
```

19:17 (Top Level) R Scri

R 4.6.1 · ~/

```
> # Create 50 data points
> X <- rnorm(50, mean = 50, sd = 10)
> Y <- 2 * X + rnorm(50, mean = 0, sd = 10)
>
> # Create hexbin object
> h <- hexbin(X, Y, xbins = 10)
>
> # Plot hexbin
> plot(h,
+      main = "Hexbin Plot of X and Y",
+      xlab = "X",
+      ylab = "Y")
>
```



Problem 24

Data: A team uses a network graph to show a simple linear process (Step 1 → Step 2 → Step 3 → Step 4)

with no branching or multiple connections.

Task: Evaluate whether a network graph was the appropriate chart type for this process.

A **network graph** is normally used to represent relationships and connections between multiple entities. It is especially useful when there are **branching paths, multiple connections, dependencies, or complex relationships**.

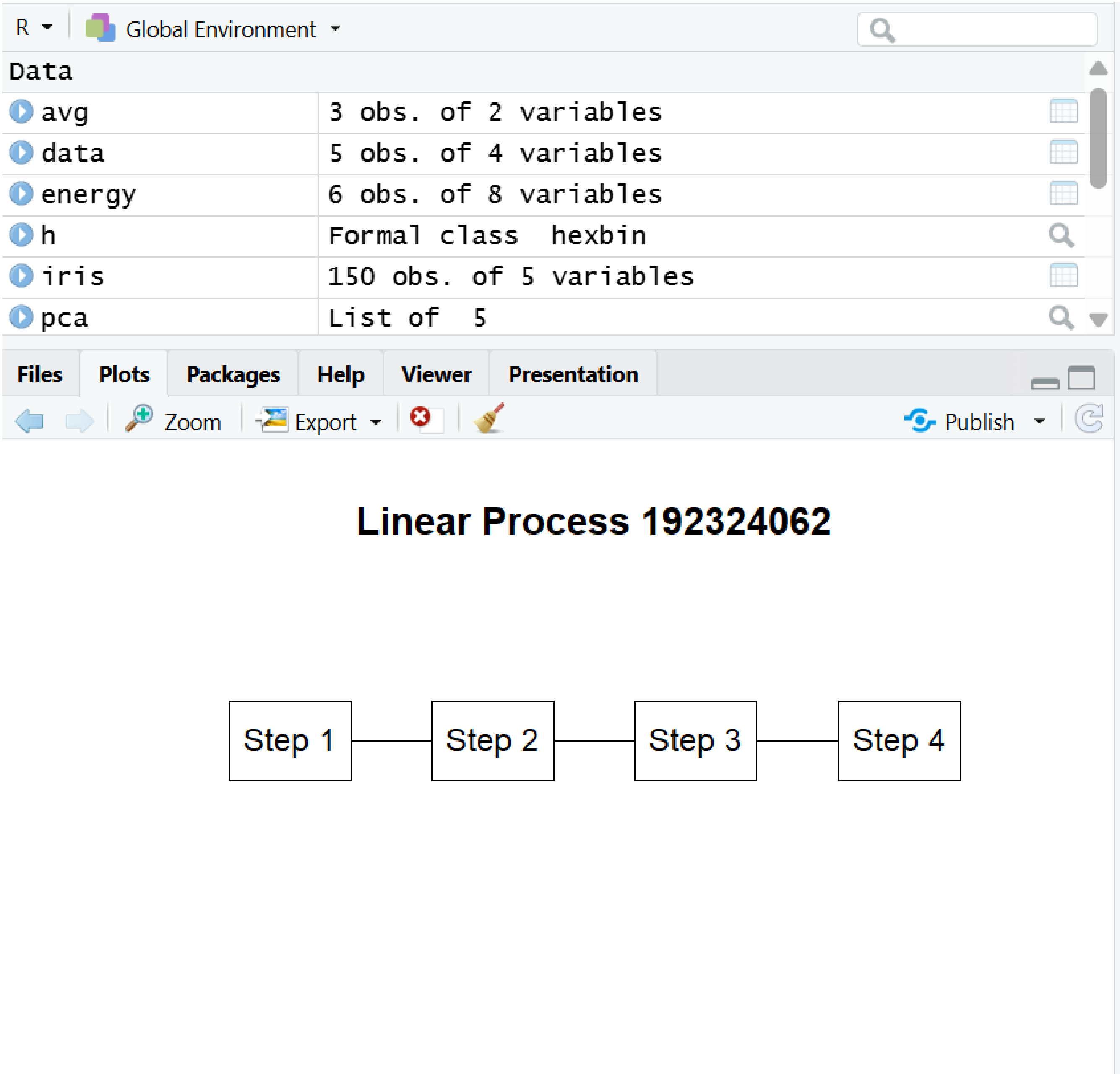
In this case, the process is simply:

Step 1 → Step 2 → Step 3 → Step 4

There is **no branching and no multiple connections**. Therefore, using a network graph is **not the most appropriate choice**.

A **flowchart** or **process diagram** would be better because it clearly shows the sequential order of the steps.

```r
# Linear Process: Step 1 -> Step 2 -> Step 3 -> Step 4

# Create a new plot
plot(1:4, rep(1, 4),
     type = "n",
     xlim = c(0.5, 4.5),
     ylim = c(0.5, 1.5),
     axes = FALSE,
     xlab = "",
     ylab = "",
     main = "Linear Process")

# Add boxes
rect(0.7, 0.85, 1.3, 1.15)
rect(1.7, 0.85, 2.3, 1.15)
rect(2.7, 0.85, 3.3, 1.15)
rect(3.7, 0.85, 4.3, 1.15)
```


R
Global Environment
Data
avg
3 obs. of 2 variables
data
5 obs. of 4 variables
energy
6 obs. of 8 variables
h
Formal class hexbin
iris
150 obs. of 5 variables
pca
List of 5
Files
Plots
Packages
Help
Viewer
Presentation
Zoom
Export
Publish
Linear Process 192324062
Step 1
Step 2
Step 3
Step 4